

DASC7011 Statistical Inference for Data Science

Tutorial 1: Review of Statistical Concepts and Get Started with R

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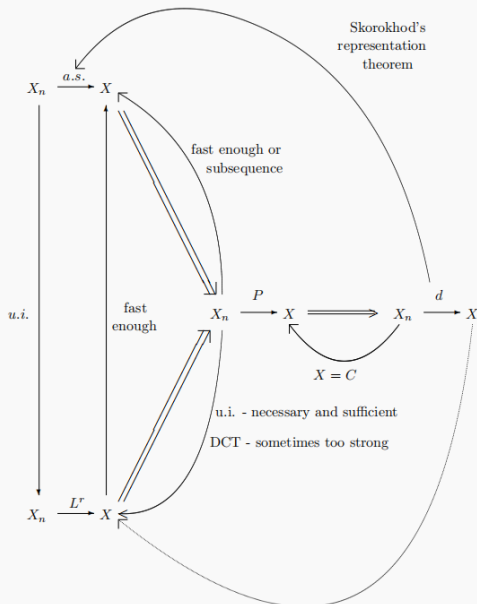
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- Convergence
- Estimation
- Credit Interval
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- Duality between Credit Interval and Hypothesis testing
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Convergence



Convergence

- **Weak Law of Large Numbers:** Let $\{X_i : i \geq 1\}$ be a sequence of i.i.d. random variables with finite mean μ , and $\bar{X}_n \triangleq \frac{1}{n} \sum_{i=1}^n X_i$. Then,

$$\bar{X}_n \xrightarrow{P} \mu, \quad \text{as } n \rightarrow \infty. \quad (1.5)$$

- **Strong Law of Large Numbers:** Let $\{X_i : i \geq 1\}$ be a sequence of i.i.d. random variables with finite mean μ , and $\bar{X}_n \triangleq \frac{1}{n} \sum_{i=1}^n X_i$. Then,

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu, \quad \text{as } n \rightarrow \infty. \quad (1.6)$$

- **Remark:** $\{X_i\}$ being i.i.d. with finite mean is the sufficient, but not necessary, condition for the convergence of $\bar{X}_n$.

- **Lindeberg-Lévy CLT:** Suppose $\{X_i : i \geq 1\}$ is a sequence of *i.i.d.* random variables with $\mathbb{E}(X_i) = \mu < \infty$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then, as n approaches infinity, the random variables $\sqrt{n}(\bar{X}_n - \mu)$ converge in distribution to a normal $N(0, \sigma^2)$:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2). \quad (1.7)$$

- The convergence in Eq. (1.7) is sometimes rewritten as

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

Definition

Estimation is the process of finding estimates for unknown characteristics of a population X . Let $\mathbf{X} = (X_1, \dots, X_n)$ be a sample from a population $X \sim M$, and $\mathbf{x} = (x_1, \dots, x_n)'$ be observations. Then, an estimator is a function of the sample $\mathbf{X}$, without any unknown factors. Such an estimator is denoted as $T = T(\mathbf{X})$.

- For example, the sample mean $\hat{\mu} = \sum_i x_i / n$ is an estimator.
- The sample variance $\hat{\sigma}^2 = \frac{\sum_i (x_i - \hat{\mu})^2}{n - 1}$ is also an estimator.

Evaluation of the estimation:

How accurate we estimate the unknown parameter θ ?

- First Moment: $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$
- Second Moment: $\text{Var}(\hat{\theta}) = E(\hat{\theta} - E(\hat{\theta}))^2$
- $\text{MSE}_{\theta}(\hat{\theta}) = E_{\theta}(\hat{\theta} - \theta)^2 = \text{Var}_{\theta}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$

Estimation

Comparison between estimators

Let $\hat{\theta}_i = T_i(\mathbf{X}), i = 1, 2$, be two estimators of θ based on the same sample $\mathbf{X}$. If for all $\theta \in \Omega$

$$MSE_{\theta}(\hat{\theta}_1) \leq MSE_{\theta}(\hat{\theta}_2)$$

then, we say that $\hat{\theta}_1$ is uniformly better than $\hat{\theta}_2$.

Efficiency

Let $\hat{\theta}_i = T_i(\mathbf{X}), i = 1, 2$, be two unbiased estimators of θ based on the same sample $\mathbf{X}$. If for all $\theta \in \Omega$

$$Var_{\theta}(\hat{\theta}_1) \leq Var_{\theta}(\hat{\theta}_2)$$

then, we say that $\hat{\theta}_1$ is uniformly more efficient than $\hat{\theta}_2$.

- The most efficient estimator (if exist), which is the unbiased estimator with the minimal variance, is called the MVUE.

Asymptotic unbiasedness

If, as n increases and tends to infinity and for all $\theta \in \Omega$,

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\theta}(\hat{\theta}_n) = \theta$$

then we say that $\hat{\theta}_n$ is asymptotically unbiased.

Consistency

An estimator $\hat{\theta}_n$ is said to be a consistent estimator of θ , if for any $\varepsilon > 0$ and for any $\theta \in \Omega$

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\theta}(|\hat{\theta}_n - \theta| > \varepsilon) = 0$$

Confidence Interval

- Let $\mathbf{X}$ be a sample from a population $X \sim \mathcal{M}(\boldsymbol{\theta})$, $\boldsymbol{\theta} \in \Omega$.
- A random and *convex* subset (a region) Ω , $C(\mathbf{X}) \subset \Omega$, is called a **credible/confidence region** at the **confidence level** $1 - \alpha$ ($0 < \alpha < 1$) if

$$\mathbb{P}_{\boldsymbol{\theta}}(\boldsymbol{\theta} \in C(\mathbf{X})) = 1 - \alpha, \quad \text{for all } \boldsymbol{\theta} \in \Omega. \quad (1.7)$$

- If $\boldsymbol{\theta} = \theta$ is a scalar character, and $C(\mathbf{X})$ has the form of an interval $(\hat{\theta}_1, \hat{\theta}_2)$, we call it a **credible/confidence interval (CI)**.
Moreover, we call
 - $\hat{\theta}_1$ the *lower credible/confidence limit (lcl)* or *lower credible/confidence bound*, and
 - $\hat{\theta}_2$ the *upper credible/confidence limit (ucl)*.

Definitions

- The null hypothesis H_0 : $\theta \in \Omega_0 \subset \Omega$, while the alternative hypothesis is H_a : $\theta \in \Omega_a \subset \Omega$ where $\Omega_0 \cap \Omega_a = \emptyset$
- Test statistic: a pivotal quantity for θ , $T = T(\mathbf{X}; \theta)$, whose distribution is conditionally known when the null hypothesis H_0 is true or marginally true.
- Hypothesis testing: Choosing a significance level (or, level of significance) α , finding a rejection region R_α (a rule) such that

$$\mathbb{P}(T(\mathbf{X}; \theta) \in R_\alpha | H_0 \text{ is true}) \leq \alpha$$

- Draw conclusion: Based on observations $\mathbf{x}$. Reject H_0 at the α -th level when $T(\mathbf{X}; \theta) \in R_\alpha$ and not reject otherwise.

Under the null hypothesis H_0 (marginally), the probability of $T(\mathbf{X}; \theta)$ taking values more extreme than, or equally extreme as $T(\mathbf{x}; \theta)$ is called the p-value of the test.

Form of R_α	Definition of p -value
$\{T(\mathbf{x}; \theta) \geq c_\alpha\}$	$\mathbb{P}_{H_0}\{T(\mathbf{X}; \theta) \geq T(\mathbf{x}; \theta)\}$
$\{T(\mathbf{x}; \theta) \leq c_\alpha\}$	$\mathbb{P}_{H_0}\{T(\mathbf{X}; \theta) \leq T(\mathbf{x}; \theta)\}$
$\{ T(\mathbf{x}; \theta) \geq c_\alpha\}$	$\mathbb{P}_{H_0}\{ T(\mathbf{X}; \theta) \geq T(\mathbf{x}; \theta)\}$

- Type I error: reject H_0 when it is true.
- Type II error: not reject H_0 when it is false.
- Power of a test: It is the probability that we correctly reject H_0 when a specific $\theta_a \in \Omega_a$ is true, that is,

$$p(\theta_a) = \mathbb{P}(T(\mathbf{X}; \theta) \in R_\alpha | \theta = \theta_a)$$

Example

Twelve standard lengths of wool were extended over a given load before and after washing. The (after-less-before) differences (cm) in the extensions were

1.7, 0.3, 0.3, 0.9, 0.4, 2.4, 1.1, 0.5, 1.2, 0.7, 0.5, 1.4

Test whether or not washing increases extensibility in general, at 5% significance using the critical region method.

Wool Extensibility Example: Results

$$H_0 : \mu_d \leq 0 \quad \text{versus} \quad H_1 : \mu_d > 0,$$

where $d = \text{after} - \text{before}$.

$$n = 12, \quad \bar{d} = 0.950, \quad s_D = 0.6446, \quad \text{SE}(\bar{d}) = 0.1861.$$

One-sample t -test

$$t = 5.1057, \quad df = 11, \quad t_{0.95,11} = 1.7959.$$

Since $5.1057 > 1.7959$, reject H_0 at the 5% level.

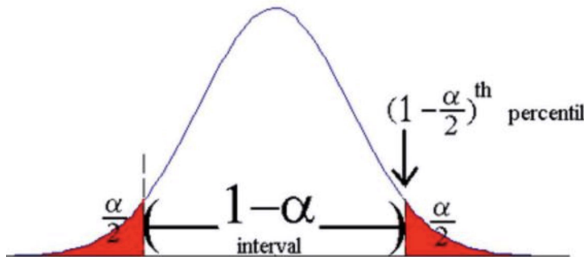
$$\text{One-sided } p\text{-value} = 0.000171.$$

Conclusion

There is strong evidence that washing increases the mean extensibility of the wool.

Duality between Confidence Interval and Hypothesis testing

- Confidence Interval: Estimation (Modelling)
Focus more on the inside part of Interval
- Hypothesis testing: Testing (Post Modelling)
Focus more on the outside part of the Interval



Duality between Credit Interval and Hypothesis testing

- For hypothesis testing $H_0 : \mu = \mu_0$ at the significance level α .

H_a	Not reject H_0 if (non-rejection region)
$\mu \neq \mu_0$	$\bar{x} \in (\mu_0 - Z_{\alpha/2}\sigma/\sqrt{n}, \mu_0 + Z_{\alpha/2}\sigma/\sqrt{n})$
$\mu < \mu_0$	$\bar{x} \in (\mu_0 - Z_{\alpha}\sigma/\sqrt{n}, +\infty)$
$\mu > \mu_0$	$\bar{x} \in (-\infty, \mu_0 + Z_{\alpha}\sigma/\sqrt{n})$

Table 1.3

- This can be rewritten into the following.

H_a	Not reject H_0 if (non-rejection region)
$\mu \neq \mu_0$	$\mu_0 \in (\bar{x} - Z_{\alpha/2}\sigma/\sqrt{n}, \bar{x} + Z_{\alpha/2}\sigma/\sqrt{n})$
$\mu < \mu_0$	$\mu_0 \in (-\infty, \bar{x} + Z_{\alpha}\sigma/\sqrt{n})$
$\mu > \mu_0$	$\mu_0 \in (\bar{x} - Z_{\alpha}\sigma/\sqrt{n}, +\infty)$

Table 1.4

Frequentist Inference

How to estimate the Bias($\hat{\theta}$) and Se($\hat{\theta}$)

Plug-in Principle:

- Find a distribution that Bias($\hat{\theta}$) or Se($\hat{\theta}$) subject to

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1) \quad (1)$$

- Use this moments (Expectation or Variance) of this distribution to build up a **link** between Bias($\hat{\theta}$) or Se($\hat{\theta}$) and the moments (Expectation or Variance) of the data

$$S^2 = \frac{\sigma^2}{n-1} \chi^2 \quad se(S^2) = \sqrt{\frac{2}{n-1}} \sigma^2 \quad (2)$$

- Plug in $\hat{\sigma}^2$ (namely S^2) for σ^2

- Example: Consider sample mean $\bar{X}$ of a normal sample $\mathbf{X} = \{X_i : 1 \leq i \leq n\} \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, where both μ and σ are unknown. Suppose we want to estimate the standard error of $\bar{X}$, namely $se(\bar{X})$.
- Sampling Distribution Theorem $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
- $se(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ with plug-in $S^2 \approx \sigma^2$
- we get $\hat{se}(\bar{X}) = \frac{S}{\sqrt{n}}$

Frequentist Inference

When there is no solution for the plug-in principle, an approximation method is needed.

Taylor approximations. Suppose that $\hat{\eta} = f(\hat{\theta})$ is a known function of an estimator $\hat{\theta}$, for which we are able to do statistical inferences. Then the (usually linear) Taylor approximation $d\hat{\eta} \approx f'(\hat{\theta})(d\hat{\theta})$ provides us a way to do inference(s) for $\hat{\eta}$, especially estimating the standard error, thinking of $f'(\hat{\theta})$ as a constant.

Installing R

R is a comprehensive statistical platform, offering all manner of data-analytic techniques. Just about any type of data analysis can be done in R.

R contains advanced statistical routines not yet available in other packages.

R has the state-of-art graphics capability. If you want to visualize complex data, R has the most comprehensive and powerful feature set available.

Installing R

- Programming language: R
- Proposed IDE: R Studio

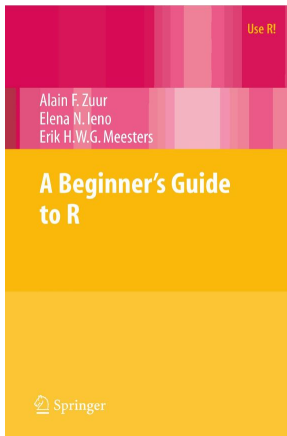


<https://www.r-project.org/>.

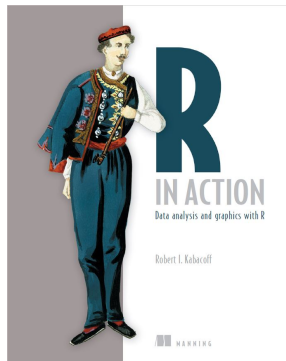


<https://www.rstudio.com/>.

Reference books for R



(a) A Beginner's Guide to R



(b) R in Action

Figure: reference books for R

Data Structures

- Vector
- Array
- Matrix
- Factor
- Data frame
- List

Basic R Programming -Working with packages

- To install a package
`install.packages("package_name")`
- To load a package
`library(package_name)`
- To get help on the use of a package
e.g. `help(package="ggplot2")`
- List all available functions with the specified text in the name
`apropos("table")`

- R in action.
Open your R Studio and start our learning in R.